

Muhammad Sohaib Ayub

Data Science Institute, University of Galway, Ireland
Email: 0133839s@universityofgalway.ie
Phone: +353 83053 8952
ORCID: 0000-0001-9206-1545
GitHub: github.com/sohaibayub

Data-driven computer scientist with expertise in machine learning, energy-efficient web technologies, sports analytics and NLP for low-resource languages. Currently a Postdoctoral Researcher at Insight Research Ireland Centre for Data Analytics, University of Galway, contributing to the EU-funded Data Spaces Support Centre. Have 10+ years of teaching experience across foundational and advanced CS courses, an active record of 28+ peer-reviewed publications, two competitive LUMS Faculty-Initiative Fund grants as Co-PI, and extensive international research collaborations. Passionate about integrating research into teaching, mentoring students in applied AI projects, and building collaborative, industry-engaged research programs.

EDUCATION

PhD Computer Science

2023

Lahore University of Management Sciences (LUMS), Lahore, Pakistan

- Thesis: “Data-Driven Approaches for Context-Aware Sports Performance Assessment and Web Energy Optimization”
- Major Courses: Big Data Analytics, Topics in Computer and Network, Network Security, Internet of Things, Multicore Computing, Human-Computer Interaction Design
- PhD Fellowship – full tuition & stipend (~PKR 7 M)
- LUMS Travel Grant to present at ICSE 2018, Sweden (~PKR 1 M)

MS Computer Science

2015

Lahore University of Management Sciences (LUMS), Lahore, Pakistan

- Thesis: “Efficiently Finding Minimal Failing Input in Map-Reduce Programs”
- Major Courses: Machine Learning, Reliable Software, Distributed Systems, Topics in Internet Research, Service Oriented Computing, Computer Architecture

BS Information Technology

2012

National University of Science and Technology (NUST), Islamabad, Pakistan

- Final Year Project: “Design and Development of Java Parallel I/O Library” [29]
- Major Courses: Web Technologies, Computing Algorithms, Operating Systems, Database Systems, Software Engineering, Computer Graphics, Data Warehousing & Data Mining, Enterprise Network Technologies (Linux)
- Fully Funded Prime Minister’s National ICT-R&D Scholarship (~PKR 1.2 M)

TEACHING EXPERIENCE

I have extensive teaching experience across multiple institutions in Pakistan. As a lead instructor at LUMS, I have enhanced student engagement through interactive teaching methods and hands-on learning approaches. In my role as co-adviser for senior year projects, I have guided more than 30 projects involving more than 150 students in software development and project management practices. My teaching experience spans from foundational pedagogy as a Teaching Assistant to developing comprehensive course materials, managing teaching teams, and providing individualized student mentorship across diverse computer science and engineering subjects.

Teaching Philosophy: I practice active, inclusive, and research-informed learning, connecting theory with real-world data. I combine scaffolded problem-solving, project-based learning, and formative feedback to nurture both technical and critical-thinking skills. I hold a Faculty Certificate in Teaching & Learning and have taught a range of introductory-to-senior CS courses to diverse cohorts. I have supervised final year projects and mentored PhD students in Software Engineering and NLP.

Institution	Course	Semester	Role
LUMS, Lahore	CS 100: Computational Problem Solving	Spring 2024	Lead Instructor
LUMS, Lahore	CS 300: Advanced Programming	Spring 2024	Lead Instructor
LUMS, Lahore	CS 100: Computational Problem Solving	Fall 2023	Lead Lab Instructor
FAST-NU, Lahore	CS 205: Operating Systems	Spring 2015	Lead Instructor
LUMS, Lahore	CS 202: Data Structures	Fall 2024	Co-instructor
LUMS, Lahore	CS 491B: Senior Year Projects Advising	Spring 2025	Co-instructor
LUMS, Lahore	CS 491A: Senior Year Projects Advising	Fall 2024	Co-instructor
LUMS, Lahore	CS 491B: Senior Year Projects Advising	Spring 2024	Co-instructor
LUMS, Lahore	CS 491A: Senior Year Projects Advising	Fall 2023	Co-instructor
LUMS, Lahore	CS 200: Introduction to Programming	Summer 2017	Lab Instructor
LUMS, Lahore	CS 510: Design and Analysis of Algorithms	Spring 2015	Teaching Assistant
LUMS, Lahore	CS 582: Distributed Systems	Fall 2014	Teaching Assistant
LUMS, Lahore	CS 310: Algorithms	Spring 2014	Teaching Assistant
NUST, Islamabad	CS 321: Enterprise Network Technologies/Linux	Spring 2012	Teaching Assistant
NUST, Islamabad	CS 251: Design and Development of Algorithms	Spring 2012	Teaching Assistant
NUST, Islamabad	ISE 312: Computing Algorithms	Spring 2011	Teaching Assistant
NUST, Islamabad	ISE 205: Object-Oriented Programming using C++	Summer 2011	Teaching Assistant
NUST, Islamabad	ISE 103: Computer Fundamentals	Fall 2010	Teaching Assistant
NUST, Islamabad	EE 221: Digital Logical Design	Spring 2012	Teaching Assistant
NUST, Islamabad	EE 241: Digital Logical Design	Fall 2011	Teaching Assistant
NUST, Islamabad	EE 221: Digital Logical Design	Spring 2011	Teaching Assistant
NUST, Islamabad	EE 241: Digital Logical Design	Spring 2011	Teaching Assistant
NUST, Islamabad	EE 241: Digital Logical Design	Fall 2010	Teaching Assistant
NUST, Islamabad	EE 241: Digital Logical Design	Spring 2010	Teaching Assistant

TEACHING & LEARNING CERTIFICATION AND ADVISING

- Universal Design Beyond the Classroom | UCD 2025
- MAP4 Postgraduate Diploma in Academic Practice | CELT, University of Galway 2025-ongoing
 - CEL6103: Research Supervision & Development
 - CEL6012: Teaching Online
- Academic Advisor, BS Students, College of Science and Engineering | University of Galway ongoing
- Faculty Certificate in Teaching and Learning: Theory and Practice | LUMS Learning Institute 2024
 - Instructional Skills Development
 - ChatGPT as Co-Instructor
 - Design Thinking for Online & In-Person Learning
 - Course Craft: Designing Effective Courses
 - From Crowd to Community: Best Practices in a Large Classroom
 - Authentic & Renewable Assessments: A Journey towards Innovative Teaching
 - Flipping and Blending Classrooms
 - Inclusive Teaching Dialogue: Beyond Relative Grading

RESEARCH EXPERIENCE

Postdoctoral Researcher

May 2025-Current

Insight Research Ireland Centre for Data Analytics, University of Galway, Ireland

- Contributing to Data Spaces Support Centre (DSSC), funded by the European Commission

PhD Research Associate

Jan 2021-Sep 2023

Data Analytics Lab, LUMS, Lahore

- Developed Predictive Modeling of Power Consumption of Websites [10] and Sports Data Analytics [16]
- 3D Wireless Sensor Networks Localization [15] in collaboration with Sun Yat-sen University, China
- Methods for Arabic OCR [14], [22] in collaboration with Islamic University of Madinah
- Synthetic Browsing Histories [9] in collaboration with Brno University of Technology, Czech Republic
- Biological Data Analysis [23] in collaboration with Department of Life Sciences @ LUMS.
- Taxi Demand Prediction [1], GenAI for contact centre business [4], Facial Emotion Detection [12], Legal Case Outcome Prediction [19], Ordinal Data Imputation [20], Global Optimization Algorithms [21] in collaboration with Massey University, New Zealand
- Mentoring PhD students from KADE Lab @ LUMS on Text Summarization [6], [8], [13], Part of speech (POS) tagging [5], [7], Sentiment Analysis [11], Cybercrime Text Analysis [17], [18].

PhD Research Fellow

Sep 2014-Dec 2020

Program Analysis Lab, LUMS, Lahore

- Verification of MPI Java Programs using Software Model Checking [25], [26]
- Efficiently Finding Minimal Failing Input in MapReduce Programs [24]

Research Assistant

Jul 2010-Sep 2012

High-Performance Computing Lab, SEECS NUST, Islamabad

- Parallel I/O Library for MPJ Express [28]
- Identification of Counterfeit Medicine using QR Codes [26]
- **Summer Research Internee 2011**
- * Conducted performance profiling of MPJ Express, identifying and addressing bottlenecks.
- **Summer Research Internee 2010**
- * Engaged in hands-on development with MPJ Express, contributing to project deliverables.
- * Proofread and edited a book on parallel computing in Java using $\LaTeX$, ensuring accuracy and clarity.

PROFESSIONAL EXPERIENCE

Software Development Engineer

Jan 2013-Aug 2013

xFlow Research Inc., Islamabad

- Developed software-defined networking applications, enhancing network management and flexibility. [27]

Software Engineer

Jul 2012-Jan 2013

Techlogix, Islamabad

- Delivered software support for Oracle FLEXCUBE, ensuring seamless operations across all Askari Bank branches.

COMMUNITY SERVICES & LEADERSHIP

- **Service to Research Community** (50+ verified peer reviews - Web of Science)
- **Journal Peer Reviews:** IEEE Transactions on Reliability, Journal of Big Data, IEEE Access, IJCDS, Jordanian Journal of Computers and IT
- **Conference Program Committee:** CHI, INTRACT, IEEE-3ICT, ADICS-ESIT, AICTC, AECT, IEEE-GCCCE, IEEE-ICOMITEE, EECIS, SCS-NCC, IEEE-ICCAIS, ICCAT, ISSPIT, ICCISN, ICSNC, ICARCAD, ISCPDM, ICST (Co-Reviewer), ISSRE (Co-Reviewer)
- **Residence Fellow, LUMS, Lahore, Pakistan**

Jun 2014-May 2017

- Managed student housing and support services part-time, fostering a supportive residential environment.
- **Volunteer for Outreach & DEI**
- Open Day, LUMS, Lahore 1st Feb, 2015
- Graduate Orientation, LUMS, Lahore 29th Aug, 2014
- Funfair for Special People, F9 Park, Islamabad 7th Apr, 2012

TALKS AND WORKSHOPS

- Presented a poster on delta debugging of MapReduce programs at ICSE, Gothenburg, Sweden 2018
- Assisted in organizing the UX Pakistan Conference at LUMS as a Student Volunteer 2017
- Facilitated a five-day Cloud Computing Workshop at LUMS as a Teaching Assistant 2015
- Delivered a talk on High-Performance Computing at Askari Bank 2013
- Participated in a one-day IBM Cloud Computing workshop at SEECS NUST 2011
- Instructed a two-day hands-on web development workshop under HL7 Student Chapter, SEECS, NUST 2011
- Supported the organization of a two-day Cloud Computing Workshop at SEECS NUST 2011
- Attended a three-day workshop on Programming Multicore Processor at HPC Lab, SEECS, NUST 2010

RESEARCH FUNDING & GRANTS

- Received PKR 1.5 million as Co-PI from the LUMS Faculty Initiative Fund in the second round
- Received PKR 1.5 million as Co-Principal Investigator from the LUMS Faculty Initiative Fund (FIF) Grant
- Received LUMS travel grant to present paper in ICSE (2018), Gothenburg, Sweden (PKR ~1m)

SELECTED AWARDS & HONORS

- Awarded LUMS PhD Fellowship covering full tuition fees and providing a monthly stipend (PKR ~7m)
- Awarded GPA-Based scholarship from SEECS NUST in Spring 2010 for outstanding academic performance
- Received fully funded scholarship (PKR ~1.2m) for BS degree from the Prime Minister's National ICT&RD Fund

SKILLS

- **Expertise:** Machine Learning, Natural Language Processing, Deep Learning, Technical Writing, Web Development
- **Programming & ML:** Python, Java, C/C++, Haskell
- **Data Processing:** Data Processing Pipelines, Big Data Analytics, SQL, Git

📄 PUBLICATIONS

- [1] S. Alam, **M. S. Ayub**, H. Cui, and M. A. Khan, "Enhancing Urban Mobility with Predictive Modeling of Taxi-Passenger Demand using Big Data Frameworks", *Public Transport*, 2025, IF: 2.4 JCR: Q1.
- [2] S. Alam, I. K. Niazi, H. Shafi, *et al.*, "Clustering-Based Balance Phenotyping in Older Adults from Treadmill Training Interventions", in *IEEE BigData*, 2025.
- [3] S. Alam, A. Pawade, **M. S. Ayub**, *et al.*, "Evaluating Explainable AI Implementation and User Agency Across Major Social Media Platforms", in *IEEE BigData*, 2025.
- [4] S. Alam, L. Skelton, **M. S. Ayub**, and W. Dharmawan, "Generative Artificial Intelligence and Digital Transformation in Contact Center Businesses", in *International Conference on Information Resources Management*, 2025.

- [5] A. Faheem, U. Faizad, **M. S. Ayub**, and A. Karim, “Roman Urdu Part of Speech (POS) Tagging”, *Natural Language Engineering*, 2025, IF: 1.9 JCR: Q1.
- [6] A. Faheem, U. Faizad, **M. S. Ayub**, and A. Karim, “Urdu Extractive Text Summarization: Evaluating Sentence Scoring Techniques”, in *Applied Intelligence and Informatics (AII) Conference*, 2025.
- [7] A. Faheem, F. Ullah, U. Azam, *et al.*, “Part of speech (POS) tagging in Roman Urdu: datasets and models”, *Language Resources and Evaluation*, pp. 1–28, 2025, IF: 1.8 JCR: Q1.
- [8] A. Faheem, F. Ullah, **M. S. Ayub**, and D. A. Karim, “Abstractive Summarization for Urdu Video Description Generation”, *ACM Transactions on Asian and Low-Resource Language Information Processing*, 2025, IF: 2.0 JCR: Q2.
- [9] D. Komosny, S. Rehman, and **M. S. Ayub**, “Synthetic Browsing Histories for 50 Countries Worldwide: Datasets for Research, Development, and Education”, *Scientific Data*, 2025, IF: 6.9 JCR: Q1.
- [10] **M. S. Ayub**, N. Mustafa, S. Ali, *et al.*, “Modeling Power Consumption of Web Apps using Static Design Features”, Under Review, 2025.
- [11] F. Ullah, S. Faizullah, I. U. Khan, *et al.*, “Prompt-Based Fine-Tuning with Multilingual Transformers for Language-Independent Sentiment Analysis”, *Scientific Reports*, 2025, IF: 3.9 JCR: Q1.
- [12] S. Alam, **M. S. Ayub**, R. Sathasivam, and M. A. Khan, “Enhanced Facial Emotion Detection Models Utilizing Geometry-Based Features for Superior Human-Computer Interaction”, in *International Conference on Neural Information Processing (ICONIP)*, 2024.
- [13] A. Faheem, F. Ullah, **M. S. Ayub**, and A. Karim, “UrduMASD: a Multimodal Abstractive Summarization Dataset for Urdu”, in *Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, ELRA, 2024.
- [14] S. Faizullah, **M. S. Ayub**, T. Alghamdi, *et al.*, “Revolutionizing Historical Document Digitization: LSTM-Enhanced OCR for Arabic Handwritten Manuscripts”, *International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 15, no. 10, 2024, IF: 0.9 JCR: Q3.
- [15] M. Islam, M. Ikram, M. Alhussein, **M. S. Ayub**, *et al.*, “A Component-Based Localization Algorithm for Sparse 3-D Wireless Sensor Networks”, *IEEE Access*, 2024, IF: 3.6; JCR: Q1.
- [16] **M. S. Ayub**, N. Ullah, S. Ali, *et al.*, “CAMP: A Player Performance Metric for Data-Driven Man of the Match Prediction in Cricket”, *Journal of the Operational Research Society*, 2024, IF: 2.7; JCR: Q1.
- [17] F. Ullah, **M. S. Ayub**, A. Faheem, *et al.*, “Towards Unveiling the Potential of Fuzzy Values as Features: A Comparative Study in Cybercrime Text Analysis”, in *International Conference on Neural Information Processing (ICONIP)*, 2024.
- [18] F. Ullah, U. Azam, A. Faheem, **M. S. Ayub**, *et al.*, “Detecting Cybercrimes in Accordance with Pakistani Law: Dataset and Evaluation using PLMs”, in *Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, ELRA, 2024.
- [19] S. Alam, R. Pande, **M. S. Ayub**, and M. A. Khan, “Towards Developing an Automated Chatbot for Predicting Legal Case Outcomes: A Deep Learning Approach”, in *Asian Conference on Intelligent Information and Database Systems (ACIIDS)*, Springer, 2023.
- [20] S. Alam, **M. S. Ayub**, S. Arora, and M. A. Khan, “An Investigation of The Imputation Techniques for Missing Values in Ordinal Data Enhancing Clustering and Classification Analysis Validity”, *Decision Analytics Journal*, vol. 9, p. 100 341, 2023, JCR: Q1.
- [21] S. Alam, X. Zhao, I. K. Niazi, **M. S. Ayub**, and M. A. Khan, “A Comparative Analysis of Global Optimization Algorithms for Surface Electromyographic Signal Onset Detection”, *Decision Analytics Journal*, vol. 8, p. 100 294, 2023, JCR: Q1.
- [22] S. Faizullah, **M. S. Ayub**, S. Hussain, and M. A. Khan, “A Survey of OCR in Arabic Language: Applications, Techniques, and Challenges”, *Applied Sciences*, vol. 13, no. 7, p. 4584, 2023, IF: 2.5, JCR: Q2.

- [23] U. Sardar, S. Ali, M. S. Ayub, *et al.*, “Sequence-Based Nanobody-Antigen Binding Prediction”, in *International Symposium on Bioinformatics Research and Applications*, Springer, 2023, pp. 227–240.
- [24] **M. S. Ayub** and J. H. Siddiqui, “Efficiently Finding Minimal Failing Input in MapReduce Programs”, in *International Conference on Software Engineering: Companion (ICSE-Companion)*, Core Rank: A*, IEEE, 2018, pp. 177–178.
- [25] **M. S. Ayub**, W. U. Rehman, and J. H. Siddiqui, “Experience Report: Verifying MPI Java Programs Using Software Model Checking”, in *International Symposium on Software Reliability Engineering (ISSRE)*, Core Rank: A, IEEE, 2017, pp. 294–304.
- [26] W. U. Rehman, **M. S. Ayub**, and J. H. Siddiqui, “Verification of MPI Java Programs using Software Model Checking”, in *ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP)*, ACM, 2016, pp. 1–2.
- [27] Q. Maqbool, **M. S. Ayub**, J. Zulfiqar, and A. Shafi, “Virtual TCAM for data Center Switches”, in *Conference on Network Function Virtualization and Software Defined Network (NFV-SDN)*, IEEE, 2015, pp. 61–66.
- [28] A. A. Awan, **M. S. Ayub**, A. Shafi, and S. Lee, “Towards Efficient Support for Parallel I/O in Java HPC”, in *International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT)*, IEEE, 2012, pp. 137–143.
- [29] **M. S. Ayub**, M. Adnan, and M. Y. Shafi, “Design and Development of a Java Parallel I/O Library”, Bachelor’s Thesis, Department of Computing, SEECS, NUST, Islamabad, Pakistan, 2012.
- [30] S. U. Rehman, R. U. Rasool, **M. S. Ayub**, *et al.*, “Reliable Identification of Counterfeit Medicine Using Camera Equipped Mobile Phones”, in *High Capacity Optical Networks and Enabling Technologies (HONET)*, IEEE, 2011, pp. 273–279.